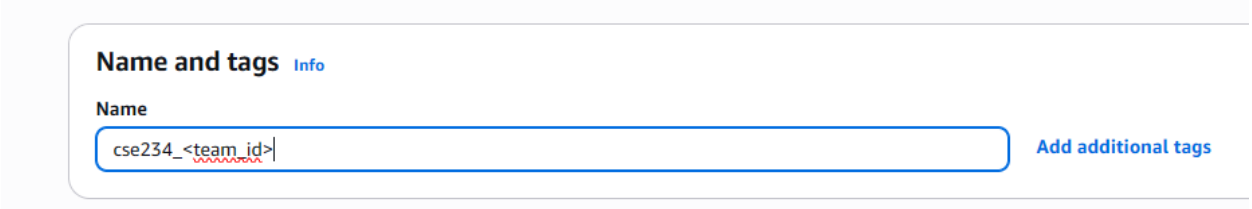
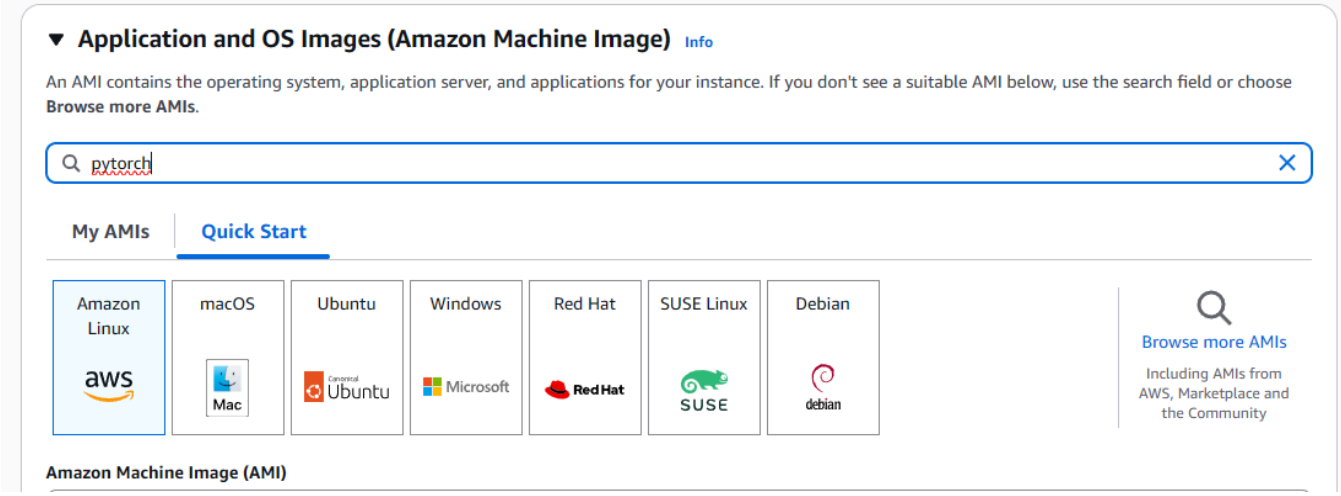
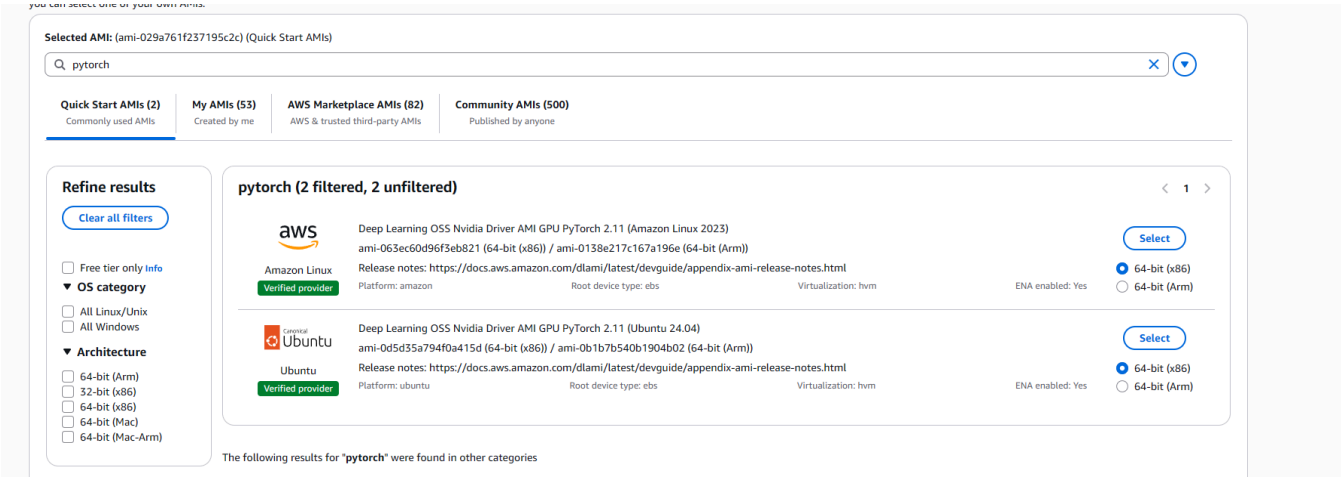
AWS Setup Instructions

- Go to https://awsed.ucsd.edu/ and click CSE234_SP26 and then choose the student role on the next page.
- This will lead you to the AWS EC2 instances dashboard. Click on Launch Instances on the right.
- Under the Names and Tags add a unique ID. Use something like the ss below

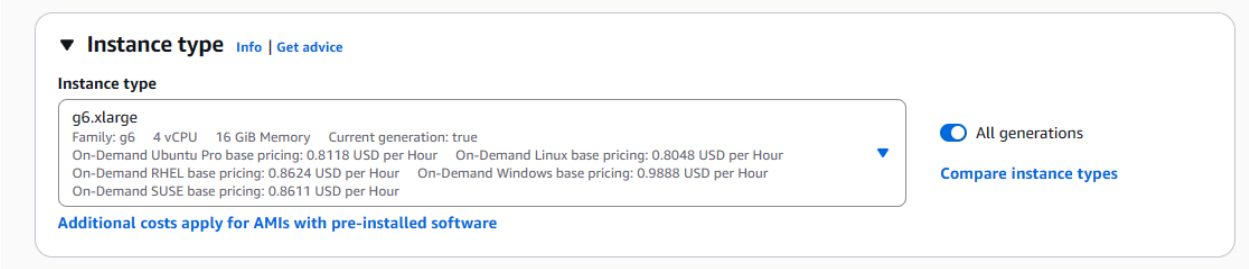


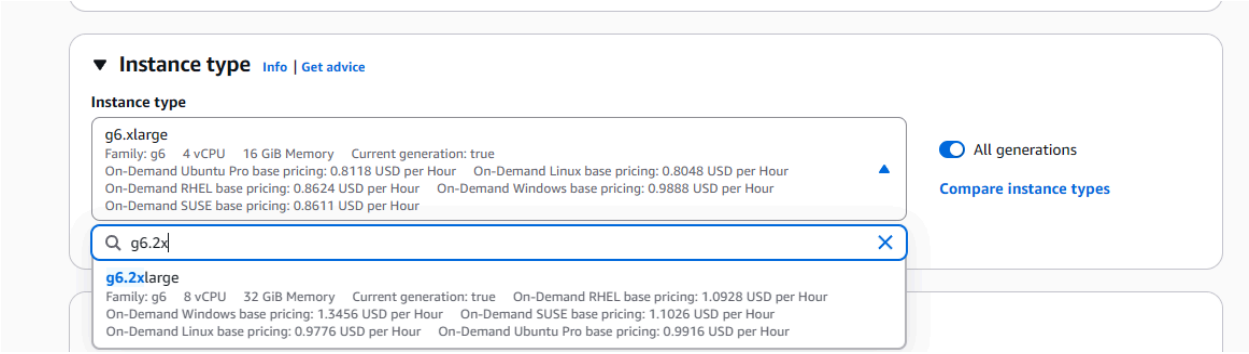
- For the AMI search for 'pytorch' and use the ubuntu (64 bit).



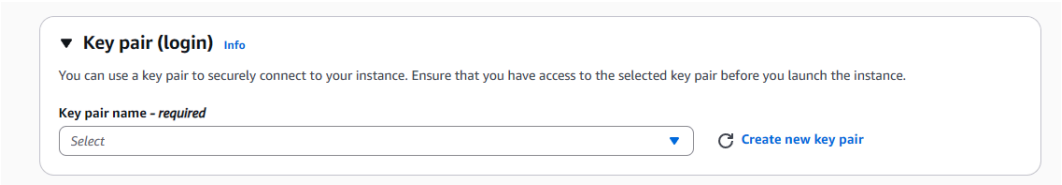


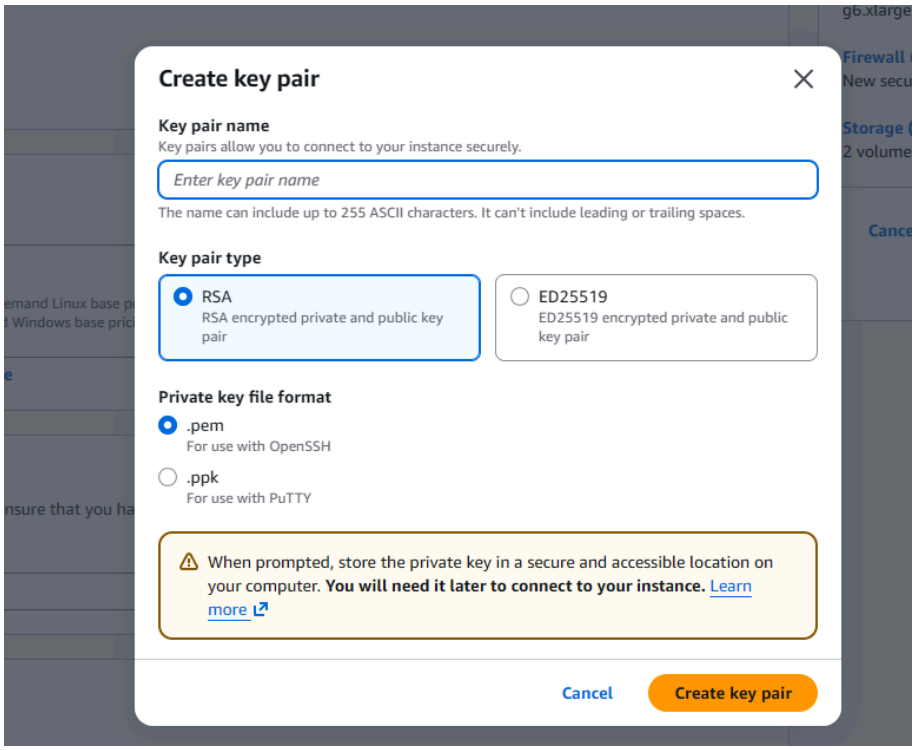
- For the instance types you can use **g6.xlarge** and **g6.2xlarge** ONLY. I would recommend using xlarge for initial debugging and use the 2x large for the final experimentations



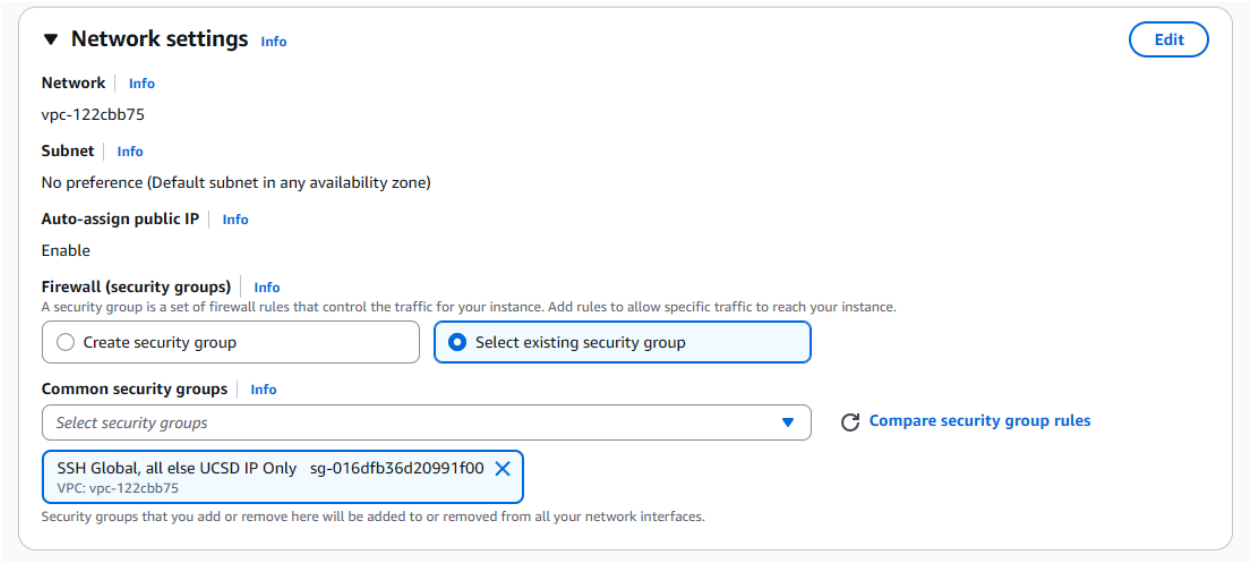


- For the Key Pair, you would have to create a new key which you will use to ssh into the instance. Create something with a unique name, which you can identify later.

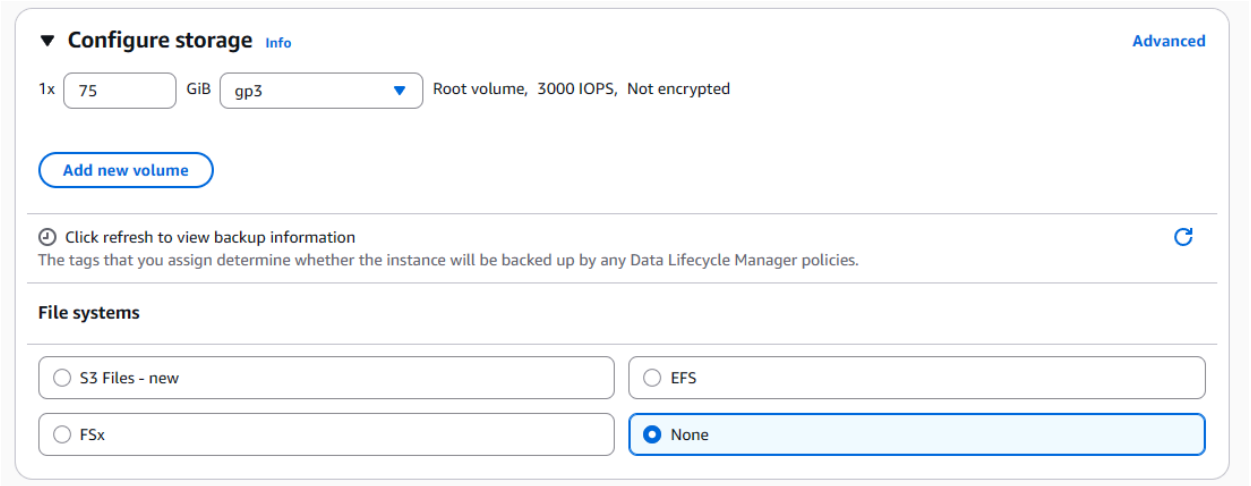




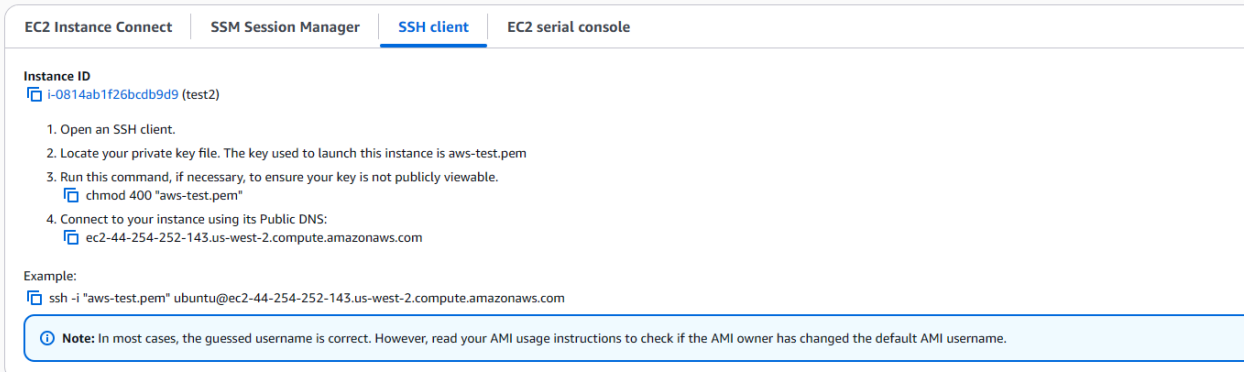
- Important - For the security groups select the existing security group and select the **SSH Global.**



- You can use 75GB max storage on your instances.



- Finally, launch the instance.
- You can look at the running instance on the EC2 instances. Click on the instance and then click connect. AWS gives you a direct command to connect. You can use the ssh command in VSCode and it will directly register the host. It should look like this -



Now you can ssh into the instance. Copy the following setup instructions into a file using

```
cat > setup.sh
```

Run using

```
bash setup.sh
```

Alternatively, you could run each command one by one.

```
source /opt/pytorch/bin/activate

python -m pip install --upgrade pip
python -m pip install uv

uv pip install \
  transformers \
  datasets \
  accelerate \
  peft \
  trl \
  bitsandbytes \
  sentencepiece \
  protobuf \
  evaluate \
  scikit-learn \
  pandas \
  loguru \
  rapidfireai

RFAI_REQ=$(python - <<'PY'
import sysconfig, pathlib
print(pathlib.Path(sysconfig.get_paths()["purelib"]) /
```

```
"setup/evals/requirements-local.txt")
PY
)




sed -i \
  -e 's/^numpy==2\.0\.1/numpy>=2.1.0/' \
  -e 's/^faiss-gpu-cu12/faiss-cpu/' \
  -e 's/^torch<=.*/torch/' \
  "$RFAI_REQ"


rapidfireai init --evals

mkdir -p ~/hf_cache ~/torch_cache ~/project

echo 'export HF_HOME=$HOME/hf_cache' >> ~/.bashrc
echo 'export HF_DATASETS_CACHE=$HOME/hf_cache/datasets' >> ~/.bashrc
echo 'export TORCH_HOME=$HOME/torch_cache' >> ~/.bashrc

source ~/.bashrc

uv cache clean
pip cache purge

python - <<'PY'
import torch
import transformers
import datasets
import peft
import trl
import rapidfireai

print("Setup complete.")
print("Torch:", torch.__version__)
print("Transformers:", transformers.__version__)
print("CUDA:", torch.cuda.is_available())
PY
```